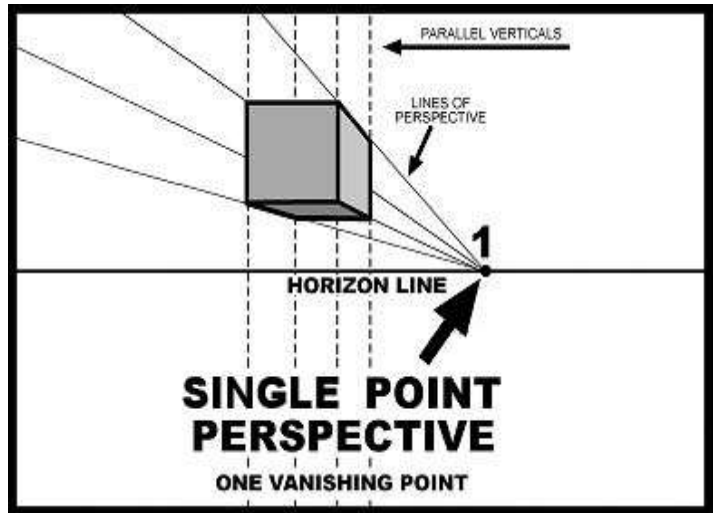




CL 343: Building Planning

Semester: 5

Student Name:

ID No:

CERTIFICATE

*This is to certify that Mr. /Ms.*_____

of _____ *Class, Roll No.* _____

Exam No. _____ *has satisfactorily completed his / her term*

work in _____

for the term ending _____ *in 20*____ */ 20*_____.

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CHANGA – 388 421

Date :

Sign of the Faculty

Head of the Department

CL 343: BUILDING PLANNING
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Tutorial 1

Conventional Signs, Symbols and Abbreviations

Symbols for Materials in Section

Sr. No.	Name of Material	Symbol
1	Brick	
2	Concrete	
3	Natural or Reconstructed Stone	
4	Partition blocks	
5	Wood	
6	Earth	
7	Plaster	
8	Glass	
9	Fibre board and Insulation board	
10	Metal Sections	

Symbols for Doors

Sr. No.	Name of Material	Symbol
1	Single Leaf Single Swing Door	
2	Single Leaf Double Swing Door	
3	Double Leaf Single Swing Door	
4	Double Leaf Double Swing Door	
5	Side Hung Door	
6	Centre Hung Door	
7	Folding Double Leaf Door	
8	Sliding Door	
9	Revolving Door	
10	Door with Projected Hinges	
11	External Rolling Shutter	
12	Internal Rolling Shutter	

Symbols for Windows

Sr. No.	Name of Material	Symbol
1	Vertical Centre Hung Window	
2	Vertical Sliding Window	
3	Horizontal Centre Hung Window	
4	Top Hung Window	
5	Bottom Hung Window	
6	Right Hand Side Hung Window	

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7	Left Hand Side Hung Window	
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Symbols for Electric Fittings

Sr. No.	Name of Material	Symbol
1	Light Plug	
2	Power Plug	
3	Meter	
4	Light Bracket	
5	Tube Light	
6	One-Way Switch	

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7	Two-Way Switch	
8	Bell Point	
9	Bell	
10	Buzzer	
11	Telephone Point	
12	Ceiling Fan	
13	Exhaust Fan	
14	Bracket Fan	

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15	Fan Regulator	
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Symbols for Plumbing Installation and Fittings

Sr. No.	Name of Material	Symbol
1	Plain Kitchen Sink	
2	Wall Lavatory Basin	
3	Corner Lavatory Basin	
4	Shower Head	
5	WC	
6	Indian type WC	
7	Stop Valve or Sluice Valve	

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8	Fire Extinguisher	
9	Fire Hydrant	
10	Gully Trap	
11	Manhole	
12	Towel Rail	
13	Rain Water Outlet	

Abbreviations

Sr. No.	Term	Abbreviation
1	Aggregate	
2	Approx	
3	Figure	
4	Diameter	
5	Not to Scale	
6	Radius	
7	Standard	
8	Height	
9	At or About	
10	Centre to Centre	
11	Floor Space Index	
12	Floor Area Ratio	
13	Mild Steel	

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14	Galvanized	
15	Galvanised Iron	
16	Cast-iron	
17	Rolled Steel Joist	
18	Column	
19	Brickwork	
20	Flooring	
21	Window	
22	Door	
23	Ventilator	
24	Finished Floor Level	
25	Ground Level	
26	Lintel Level	
27	Sill Level	
28	Plinth Level	

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29	Reinforced Cement Concrete	
30	Plain Cement Concrete	
31	Brick Bat Cement Concrete	
32	Cement Mortar	
33	Lime Mortar	
34	Damp Proof Course	
35	Floor Trap	
36	Full Supply Level	
37	Inspection Chamber	
38	Manhole	
39	Water Closet	
40	Floor Trap	

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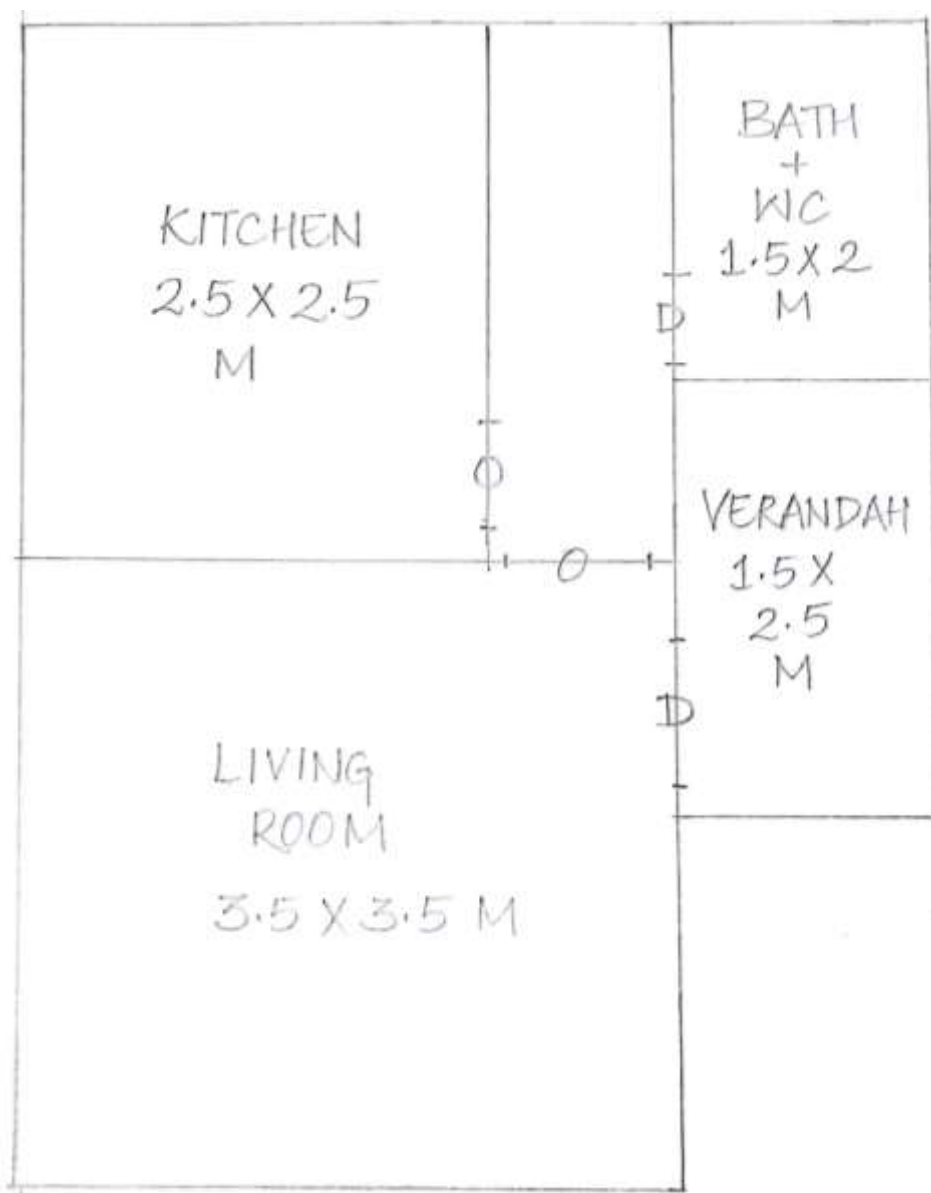
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Tutorial 2

Plan of a Simple Residential Building

Objective: Creating a plan of a single storied simple Residential building from the given line diagram. Scale -1:50.

Examples:



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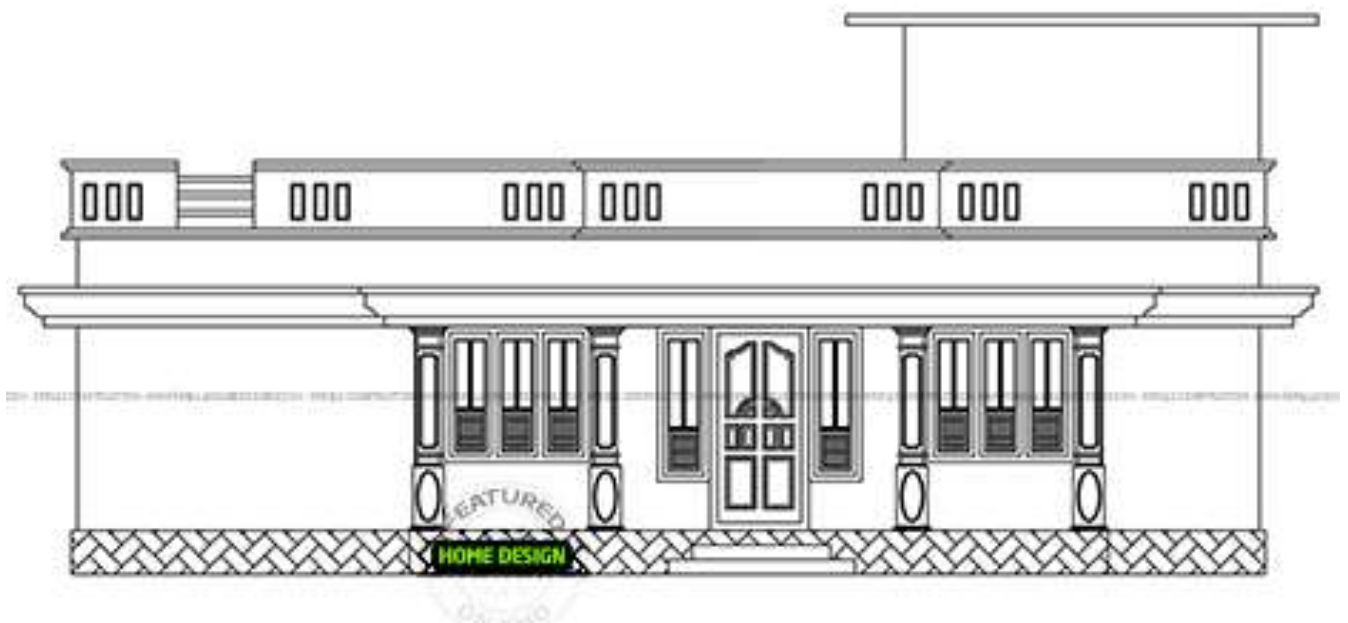
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Tutorial 3

Elevation of a Simple Residential Building

Objective: Creating an elevation of a single storied simple Residential building from the given plan. Scale -1:50.

Examples:



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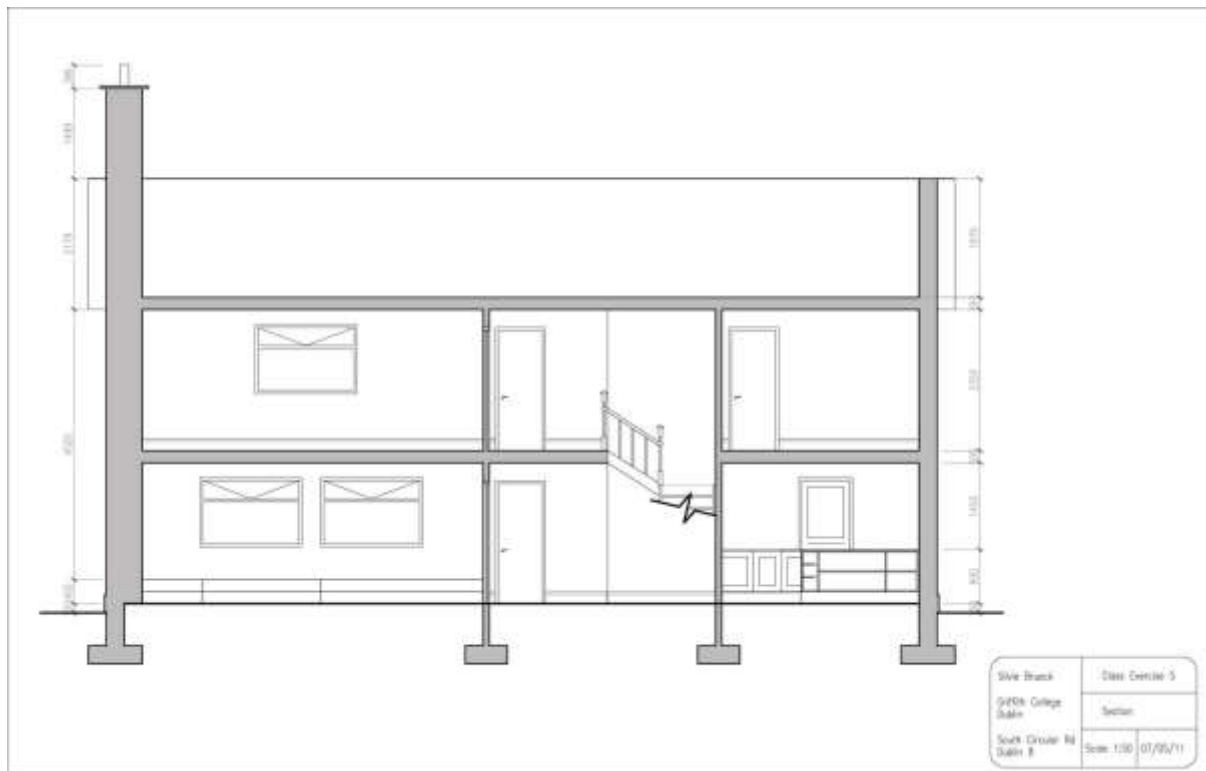
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Tutorial 4

Section of a Simple Residential Building

Objective: Creating a section of a single storied simple Residential building from the given plan. Scale -1:50.

Examples:



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Tutorial 5

Plan of a Duplex Residential Building

Objective: Creating a ground floor plan, first floor plan and terrace floor plan of a two storied duplex residential building from measure drawing of a house taken by student. Scale -1:50 or any other suitable scale.

Examples:



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Tutorial 6

Elevation of a Duplex Residential Building

Objective: Creating an elevation of a two storied duplex residential building from the ground floor plan, second floor plan and terrace floor plan prepared in tutorial 5. Scale -1:50 or any other suitable scale.

Examples:



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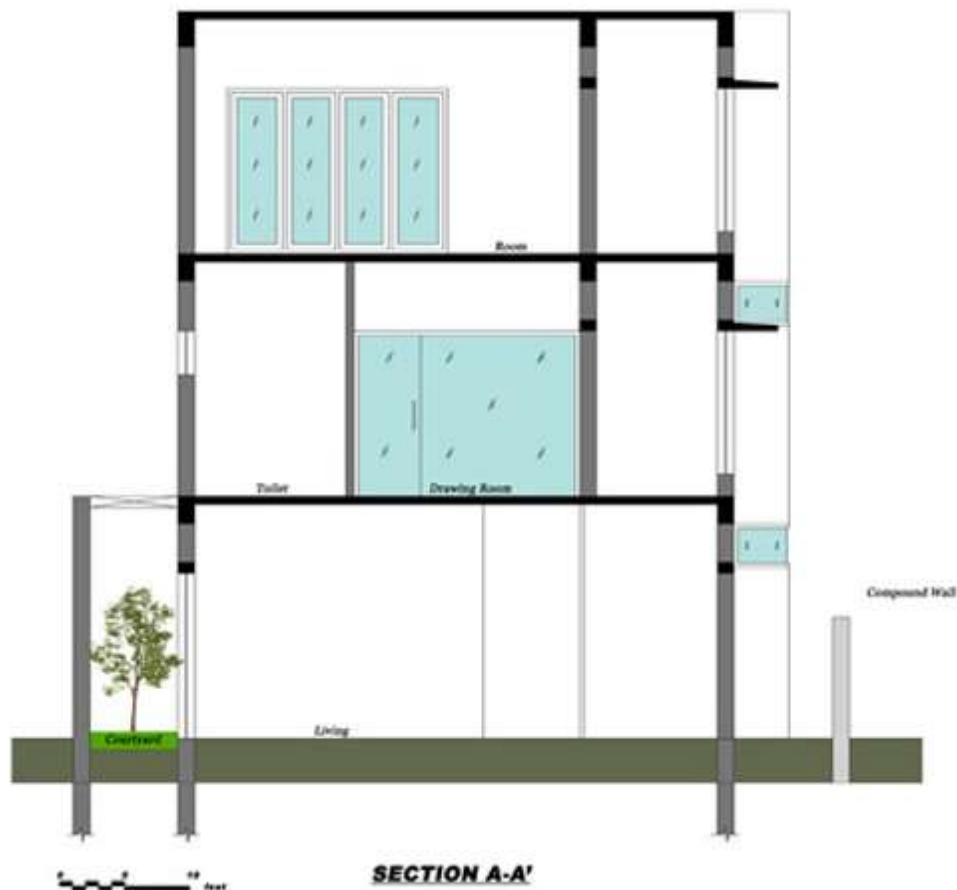
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Tutorial 7

Section of a Duplex Residential Building

Objective: Creating a section of a two storied duplex residential building from the ground floor plan, second floor plan and terrace floor plan prepared in tutorial 5. Scale -1:50 or any other suitable scale.

Examples:



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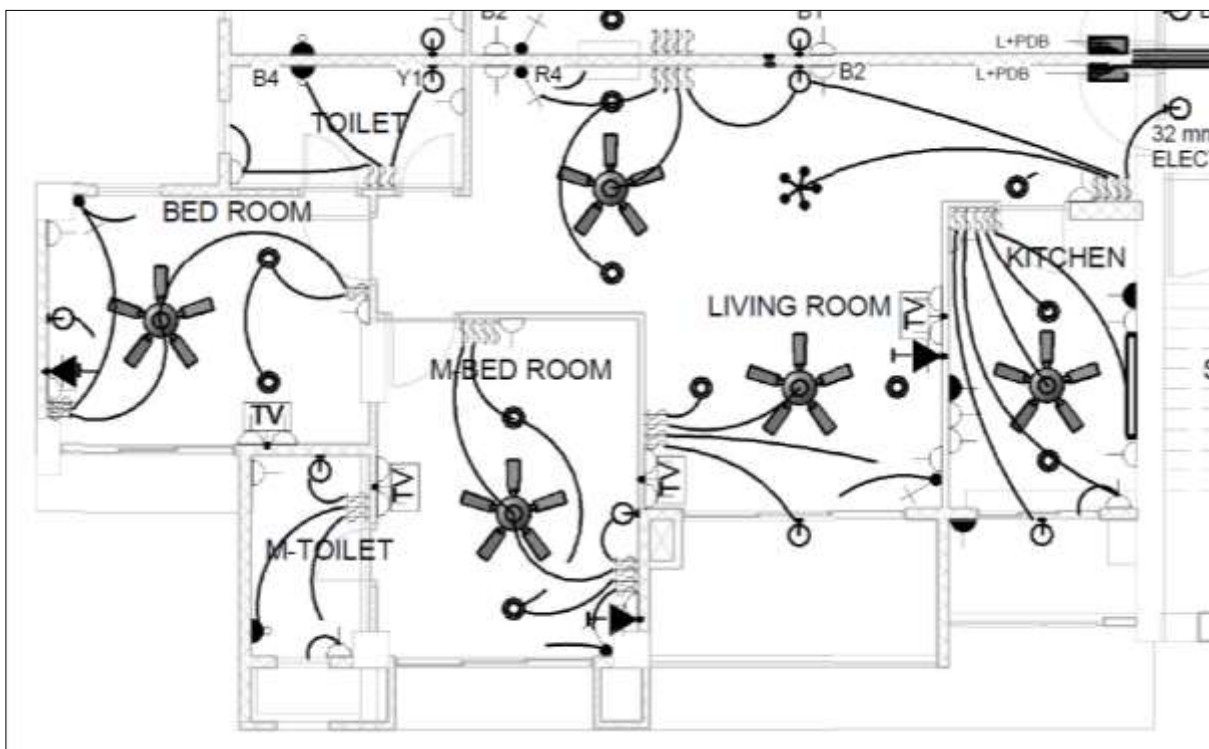
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Tutorial 8

Electrical Layout

Objective: Give the electrical details in the ground floor plan drawn in tutorial 5.

Examples:

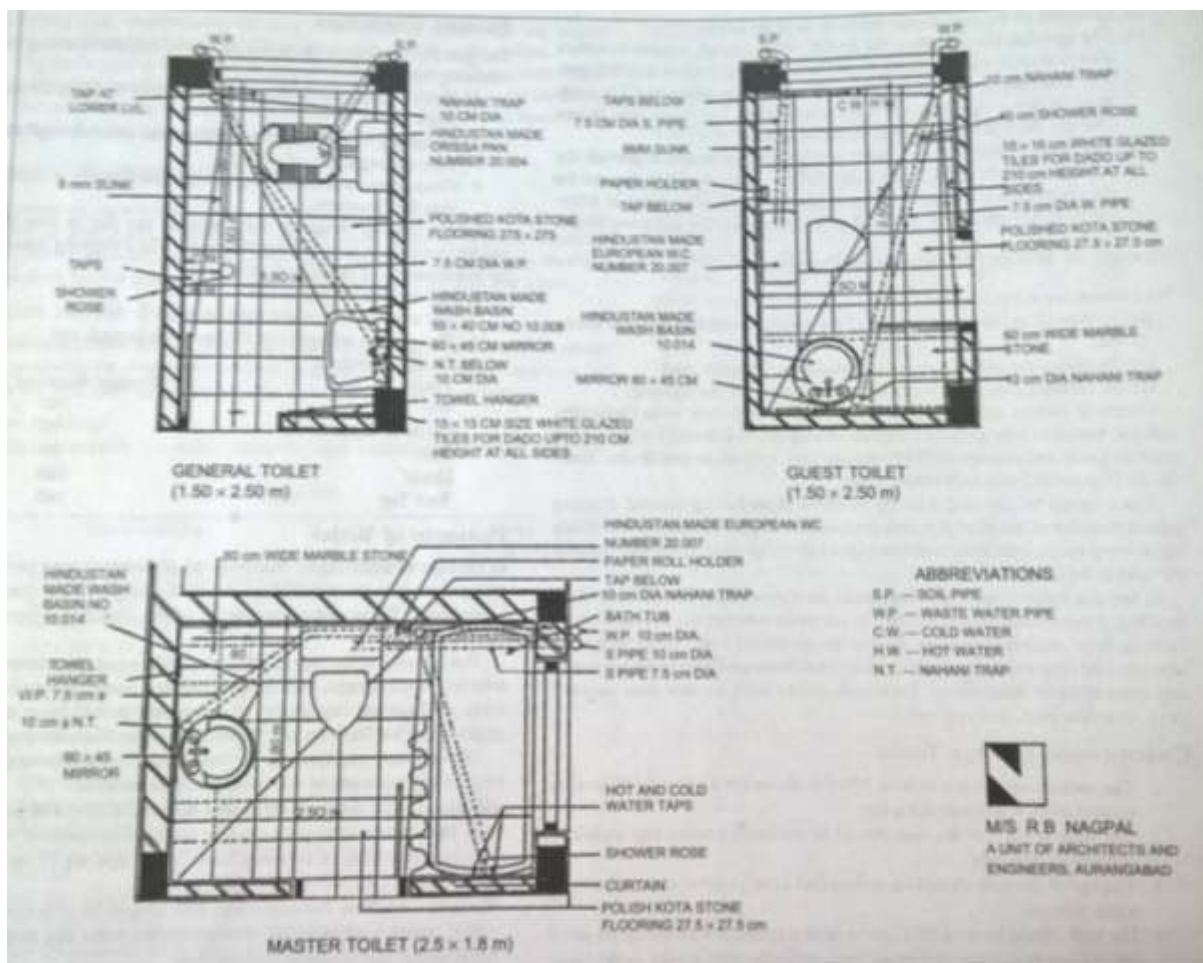


Electrical Layout

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Plumbing Layout

Examples:



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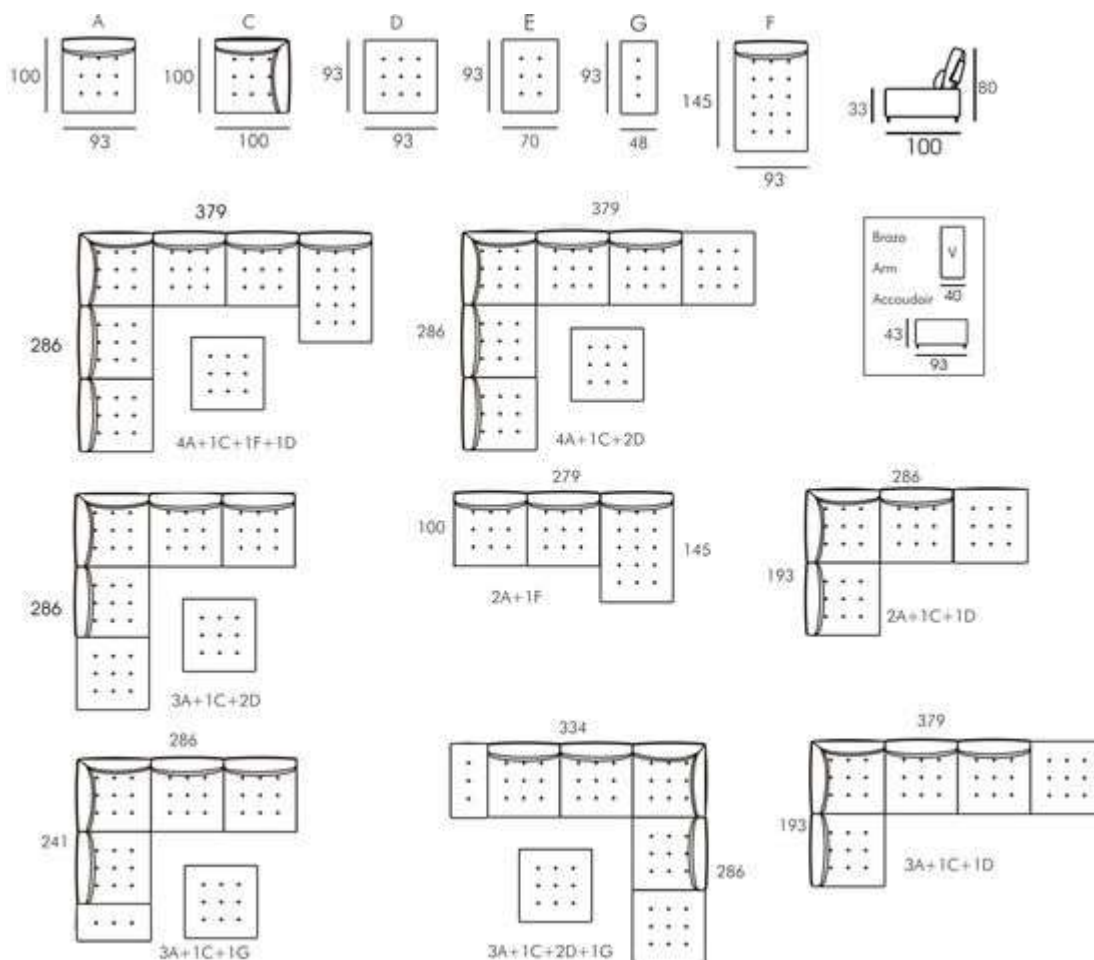
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Tutorial 10

Furniture Layout

Objective: Give the furniture details in the ground floor plan drawn in tutorial 5.

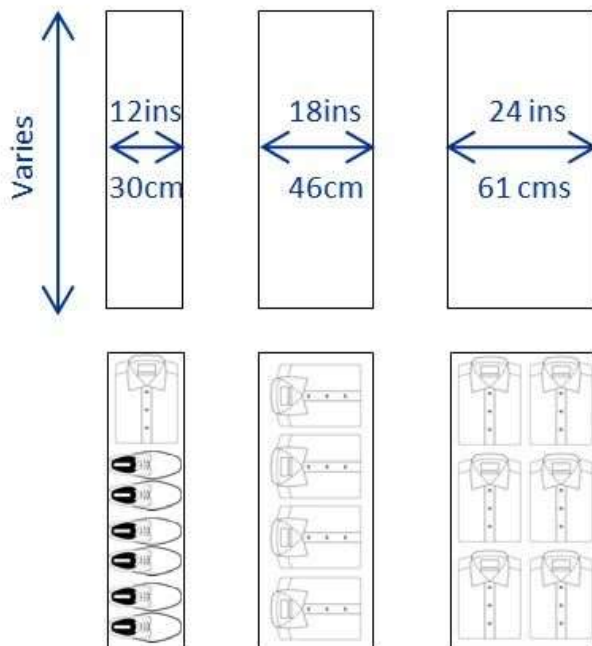
Examples:



Dimensions for Sofa



Dimensions for Bed



**Dimensions for Wardrobe
(Width only)**

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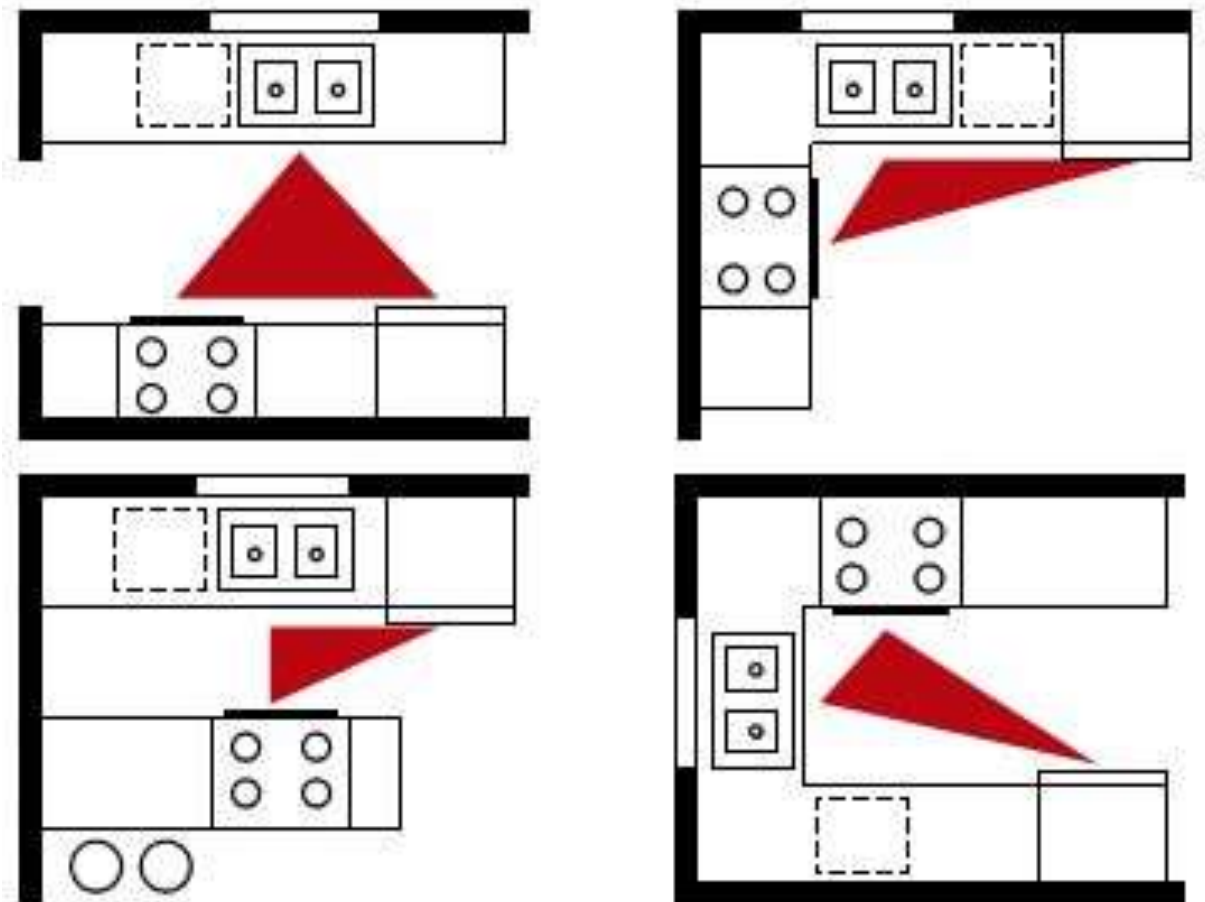
Tutorial 11

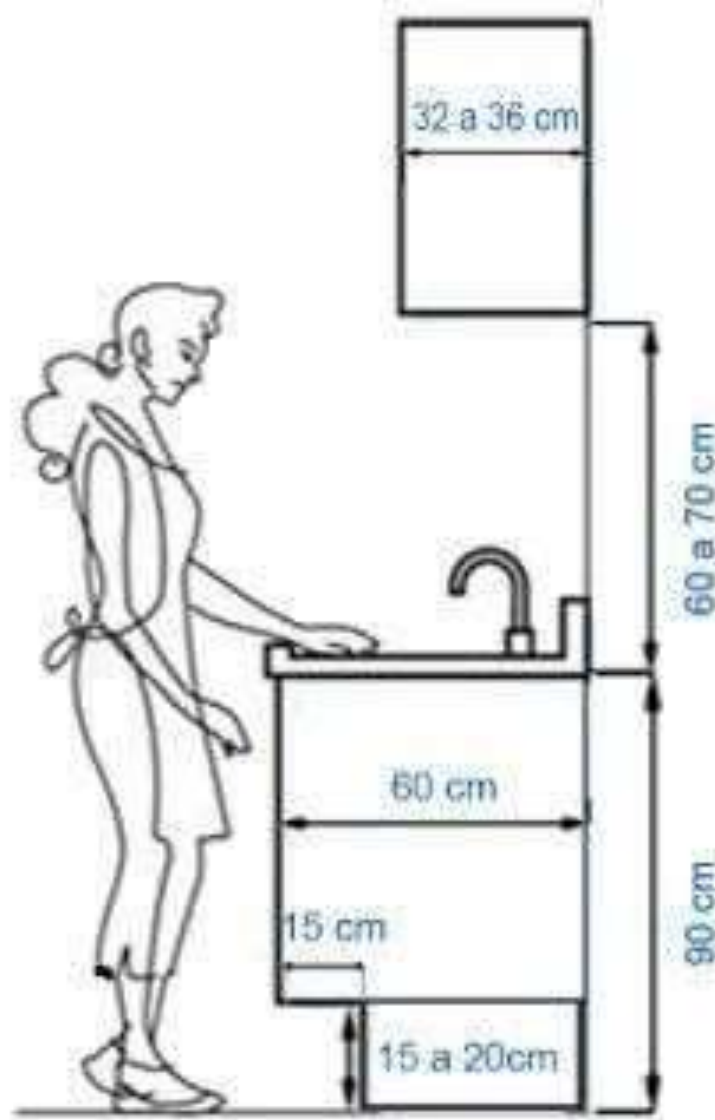
Kitchen Layout

Objective: Give the kitchen layout details in the ground floor plan drawn in tutorial 5.

Examples:

Kitchen Triangle





Dimensions for Platform

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Tutorial 12

Design of a Residential Building

Objective: Planning of a Two Storied Residential building – Plans, Section, Elevation, Section, Lay-out, Schedule of Openings, Scale -1:50

Requirements

- Plot area – 15 x 18 m
- Climate conditions – Hot Dry Climate
- FSI – 0.9
- Allowable built up area on Ground level – 65%
- The shorter side of the plot faces the road in North Direction.

Ground Floor

- Drawing room
- Bed room
- Dining area
- Staircase leading to the first floor from the dining area
- Kitchen with store attached
- One common toilet

First Floor

- Master bed room with attached toilet
- Bed room with attached toilet
- 1 balcony

Building Bye-laws

Different types of Residential Buildings plot sizes and their frontage

Type of Residential Building	Plot size m ²	Frontage m
Detached Building	Above 250	Above 12
Semi-detached building	125-250	8 to 12
Row type building	50-125	4.5 to 8

Open Space towards the Width of Street

Width of street fronting the plot (m)	Front open space minimum (m)
Upto 7.5*	1.5*
7.5 to 18	3.0
18 to 30	4.5
Above 30	6.0

Side Open Spaces based on the height of the building

Height of Building (m)	Side and Rear open spaces to be left around the building
10	3
15	5
18	6
21	7
24	8
30	10
40	12
50	14
53 and above	16

Minimum Size and Other Requirements of Different Types of Rooms

ROOM TYPE	MIN AREA (SQ.M)	MIN SIZE OF		OTHER REQUIREMENTS
		Side (m)	Height (m)	
Habitable room (bed/living/drawing /dining/study room)				
where there is only one room	9.5	2.4	2.75 (surface of floor to lowest point of ceiling)	Min 2.4 m headroom, in case of AC duct/ false ceiling
if more than one room	9.5 for one & 7.5 for other	2.4 & 2.1 resp.		
Kitchen	4.5	1.8	2.75	Floor should be impermeable. Unless separate pantry provided, means for washing kitchen utensils which shall lead directly or through a sink to grated and trapped connection
Kitchen + store	5.0	1.8	2.75	

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Kitchen + store + dining area	7.5	2.1	2.75	to waste pipe
Bath room	1.8		2.0	At least one wall should open to external air. Should not be directly over or under any room other than another latrine/washing place/bath unless the floor is water tight. Platform shall be made of non-absorbent materials. Floor should be impermeable with slope towards drain.
Water-closets (w/c)	1.1		2.0	
Bath + w/c	2.8	1.2	2.0	
Store room	3	-	2.2	-
Garage	12.5	-	2.4	-
Stair case		Min width of stair 1.0 m	Min clear head room shall be 2.2 m	For residential building, Min Tread (w/o nosing) = 250 mm Max Rise = 190mm

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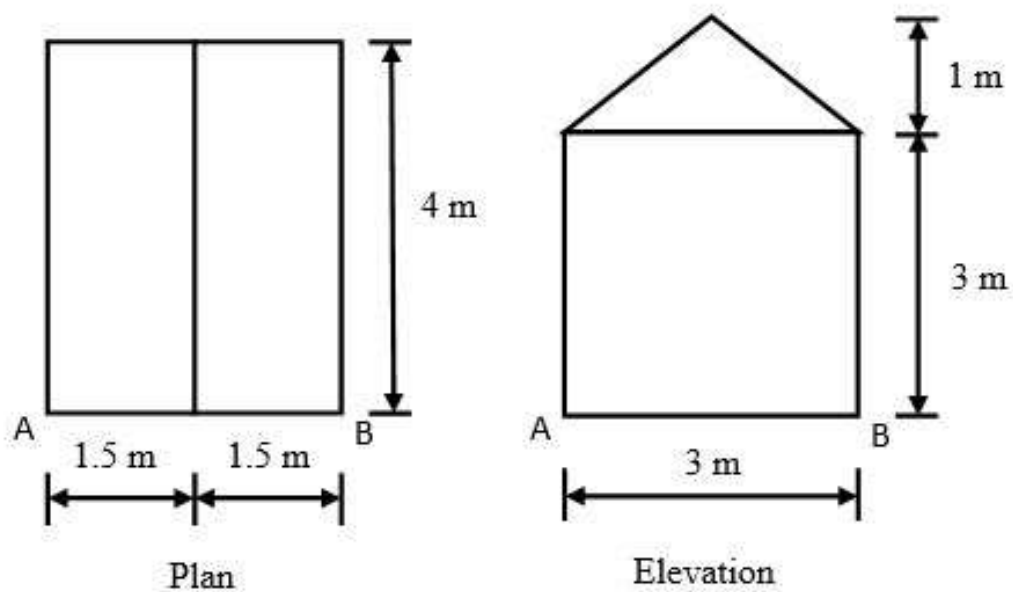
Tutorial 13

One Point Perspective Drawing

Objective: To draw a one point perspective view for the drawing given.

Details

- Scale to be taken – 1:50 or more.
- The base is on the ground plane and the vertical edge between points A and B in the picture plane.
- The station point is 2 m in front of the picture plane, 3 m from the right side of point B and 2.5 m above the ground plane.



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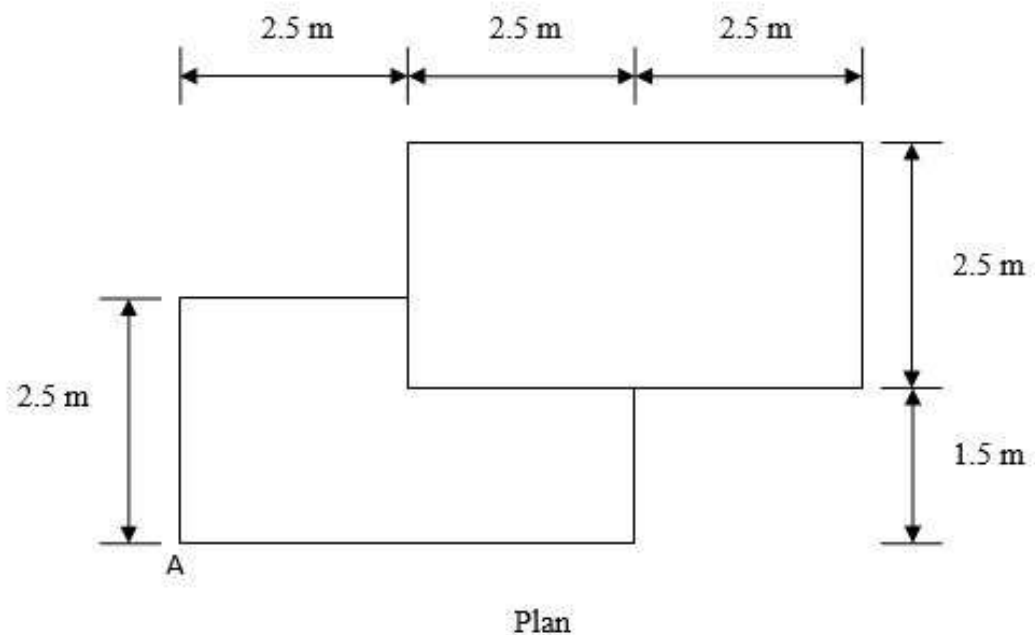
Tutorial 14

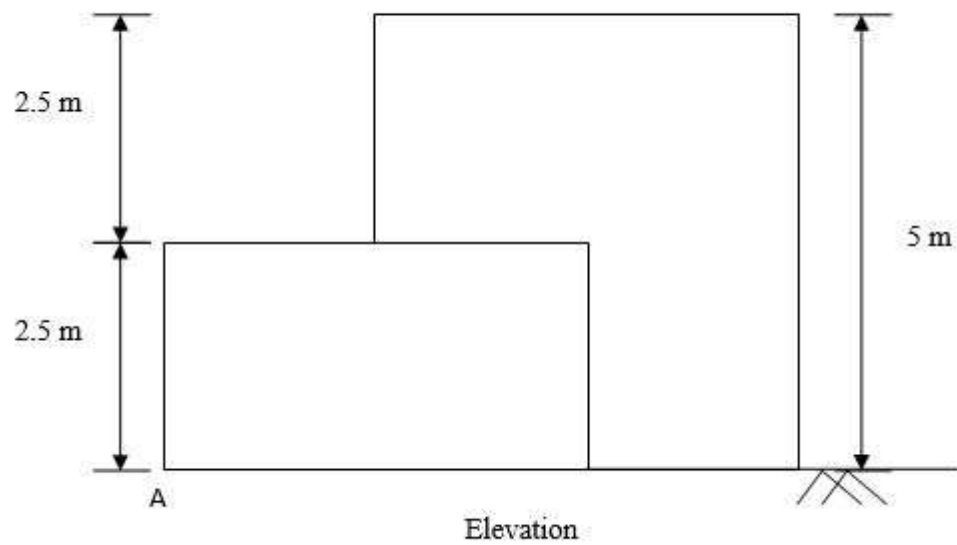
Two Point Perspective Drawing

Objective: To draw a two point perspective view for the drawing given.

Details

- Scale to be taken – 1:50 or more.
- The base is on the ground plane and vertical edge at point A in the picture plane. The station point is 3 m in front of the picture plane and 3 m above the ground plane.





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Tutorial 15

Building Bye-laws

Q1	Write a short note on bye-laws regarding 'Height of the Building Limitations'
Q2	Write a short note on byelaws for lighting and ventilation.
Q3	Define Floor Area Ratio. Explain how FAR helps in controlling the development activity in a town?
Q4	A plot owner proposed G + 1 construction with 175 m ² built up area on each floor, on a plot of size 16 m x 21 m. If all side margins are 2 m, find the ground coverage and FSI proposed. If the FSI allowed in the area is 1.0, state with reasons whether the plan will be sanctioned or not.
Q5	A plot is having dimensions 30 m x 40 m. A building constructed on it occupies 400 m ² on ground and 380 m ² on first floor. If the permissible FSI is 0.8, how much area can be constructed on the second floor?
Q6	A plot owner proposed G + 1 construction with 225 m ² on each floor, on a plot of size 20 m x 25 m. If all margins are 2.5 m and F.S.I allowed is 1.0, calculate the additional construction possible in plot. If maximum possible construction is to be made on the ground floor after sparing the margins, calculate the possible additional construction on the ground and first floor if any.

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Tutorial 16

Town Planning

Q1	Define: Town Planning, Horizontal growth, Vertical growth
Q2	Classify Ancient towns and explain any one in detail.
Q3	List out necessity of Town Planning.
Q4	Explain “Principles of Town Planning” in detail.
Q5	Classify the “Growth of Towns”
Q6	Write a short note on: Concentric growth, Ribbon Development and Satellite Town.

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Tutorial 17

Model Making

Objective: To facilitate understanding of the different components of building design by constructing a G + 1 model of a residential building.

Criteria and Rules for Model Making Project

- Model must be a residential building.
- Scale of the model - 1:25 (i.e. 1m = 4 cm).
- Model must be made by students at home and brought on the day of the competition.
- Model can be made from the following materials- (cardboard, mount board or any other suitable material, except thermocol)
- Team shall consist of not more than 5 members (Class wise)
- Every group will be given a specific plot size, FSI, maximum allowable built up area on GF, margins, directions etc. on the basis of which they have to design the building.
- The model should be aesthetically pleasing.
- Slab in the model should be removable from the top so the details of the building can be seen.

NOTE: More Criteria, Rules and Topics are given by respective faculty during the tutorial sessions.

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